Hardware Requirements Specification

for "Arduino's Moving Castle"

Version 1.0

28/09/2025

Table of Contents

1. Project Overview —— **3**

1.1 Background and Audience ---------- 3

1.2 Purpose and Scope —---------- 3

1.3 Project Timeline ---------- 3

2. Product Description —---------- **4**

2.1 Product Context and User Characteristics ---------- 4

2.2 Assumptions and Risks ---------- 4

2.3 Contraints —---------- 5

2.4 Dependencies —---------- 5

3. Requirements —---------- **5**

3.1 Functional Requirements —---------- 5

3.2 User Interface Requirements ---------- 6

3.3 Hardware Interface Requirements ---------- 6

3.4 Hardware Component Requirements ---------- 7

3.5 Visual and Design Requirements ---------- 8

Appendix ———————————————————————————————— **9**

A Definitions, Acronyms, and Abbreviations —---------- 9

B Concept Art and Photos ---------- 10

C Design Inspiration —---------- 11

1. Project Overview

1.1 Background

Arduino's Moving Castle is a functional, miniature recreation of Howl's Moving Castle, brought together with Arduino components. With a unique blend of fantasy and technology, this project will use programmable hardware to create a whimsical castle full of life and motion. The castle will imitate natural movements, with complex walking sequences, LED lights, and the tune of the iconic OST from the movie, "Merry go round of life", playing in the background. This project is designed to appeal to both makers and engineers alike, integrating both visual art and technology.

1.2 Purpose and Scope

The purpose of this specification is to outline the development cycle and delivery requirements of this project. This document will serve as the planning guide to future development for *Arduino's Moving Castle* by defining the project and its technical requirements. It will ensure that the final product meets both curriculum expectations set by the manager, and client requirements.

The final product will be a miniature recreation of HMC, by Studio Ghibli. This product is not meant for commercial use, but instead as a creative showcase in an academic setting.

1.3 Project Timeline

- 28/09/2025: Finish Hardware Requirements Specification Document (HRS)
- 24/10/2025: Finish Preliminary Design Document (PDD)
- 22/12/2025: Finish body hardware (walk cycle) and Software
- 10/01/2026: Finish Body testing and solder
- 23/01/2026: Finish building the Castle

2. Product Description

2.1 Product Context and User Characteristics

Arduino's Moving Castle is a self-contained system, able to function independently using arduino-based components and software. It doesn't require external systems during operation.

User Profile 1:

- Student/Teenager: basic technical experience with electrical components
- Likely to interact with the castle with curiosity during development.

User Profile 2:

- Teacher: highly experienced with robotics and arduino platform.
- Will interact with the castle for academic assessments.
- Will have received detailed information about the castle.

2.2 Assumptions and Risks

- Designer Experience:
  - The designer has limited experience with CAD for 3d printing
  - The designer has basic knowledge of Arduino and software, and will be using some components for the first time.
  - The designer has never created a miniature castle model before, final aesthetics may be limited.
- Component availability:
  - Many components will be purchased online, research will be needed to identify quality components for a reasonable price that fulfill the requirements.
- Component Compatibility:
  - Assume all components bought can run on a 5V power source.

2.3 Constraints

- Arduino on-board memory
- Component and material costs
- Maximum weight capacity of the body
  - Controls the final size and materials of the castle, testing required.

2.4 Dependencies

- The legs need to be 3d printed before the servo motors can be attached.
- The base of the castle needs to be printed before the legs can be fastened.
- Wires need to be measured, crimped, and created before the legs can be attached.
- The body of the castle needs to be built and tested on breadboard before a shield can be soldered
- A functional base of the castle needs to be walking before it can be tested for weight.
    - This will dictate the size of the decorative castle.
- The body of the castle needs to be able to walk before the ultrasonic sensor can be tuned

3. Requirements

3.1 Functional Requirements

Req #	Requirement	Comments	Date Reviewed	Customer
HRS-3.1.1	The body of the castle will be able to carry over 500g of payload.	The decorative castle will have to be supported by the body.		
HRS-3.1.2	The legs of the castle will all be able to move independently.	The castle will have a smooth walk cycle, enabled by the independent joints.		
HRS-3.1.3	The castle will automatically stand when turned on.	The basic off positioning of the castle will be with the legs all bent and tucked behind, and the base resting on the ground.		
HRS-3.1.4	The castle will be able to change directions	The castle will be able to turn left and right along a predetermined path.		
HRS-3.1.5	*optional* The castle will be able to detect and respond to obstacles 3 castle steps in front of it.	The length of one castle step will be determined during testing. An oscilloscope will be used to detect obstacles.		

HRS-3.1.6	*optional* The castle will be able to play the song "merry go round of life"	It's the OST of Howl's Moving Castle.		

3.2 User Interface Requirements

Req #	Requirement	Comments	Date Reviewed	Customer Review
HRS-3.2.1	The command line will be able to control each joint in the legs individually when plugged into the computer.	For testing purposes, the command line will be able to override the Legs. (hardware interrupts)		
HRS-3.2.2	*optional* The direction of the castle will be controlled by joystick			
HRS-3.2.3	*optional* The direction of the castle will be controlled by wasd keys in the serial monitor.	For testing, control the castle when plugged in through the Computer.		

3.3 Hardware Interface Requirements

Req #	Requirement	Comments	Date Reviewed	Customer Review
HRS-3.3.1	The Microcontroller will drive the servo motor driver, which shall then control the 12 servo motors			
HRS-3.3.2	The Ultrasonic sensor will use Trigger and Echo pins to send and receive a sound burst.	The time it takes for the burst to get there and back will then be translated to distance.		

3.4 Hardware Component Requirements

Req #	Requirement	Comments	Date Reviewed	Customer Review
HRS-3.4.1	The legs will need 12 servo motors, 3 in each leg.	The servos need to be powerful enough to carry 500g together, and take in 5v.		
HRS-3.4.2	The castle will be powered by batteries.	Batteries allow the castle freedom, so it's not tied down to a computer.		
HRS-3.4.3	The castle will use a shift register to increase the number of digital outputs available.			
HRS-3.4.4	*optional* The castle will have a memorySD card that it can read step sequences from.	Memory dense processes like music track and step sequences and be stored here to save space in the arduino.		
HRS-3.4.5	*optional* The OST of HMC will be played by a passive buzzer.			
HRS-3.4.6	*optional* Red LEDs will be placed near the heart of the castle to represent Calcifer.			

3.5 Visual and Design Requirements

Req #	Requirement	Comments	Date Reviewed	Customer Review
HRS-3.5.1	The material used for the castle will be a gradient, with heaviest at the bottom, and lightest at the top.	**Light**: styrofoam, paper/cardboard **Medium**: 3d printed pieces, lego **Heavy**: plastics, popsicle sticks		
HRS-3.5.2	The castle will weigh less than 500g.			
HRS-3.5.3	The castle will be based on the castle from HMC			
HRS-3.5.4	The castle will be detachable from the top.	The top will come off for easy access to the hardware components.		
HRS-3.5.5	The legs and body of the castle will be 3d printed.	All 3d printed parts will be sanded and painted.		
HRS-3.5.6	The main components will be soldered together.	After the castle is functional on a bread board, the pieces will be soldered together.		
HRS-3.5.7	*optional* Wires will be crimped, fitted, and painted to blend into the castle.	The wires will be both functional and decorative.		

Appendix

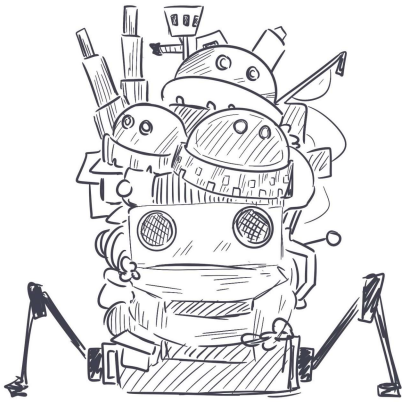
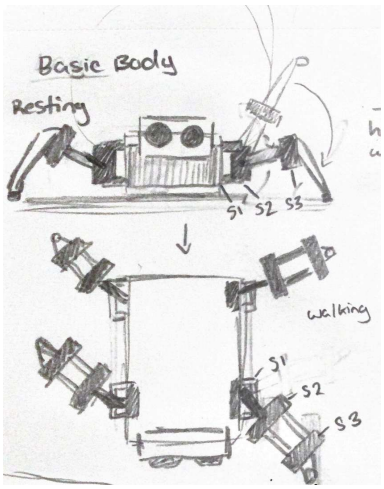
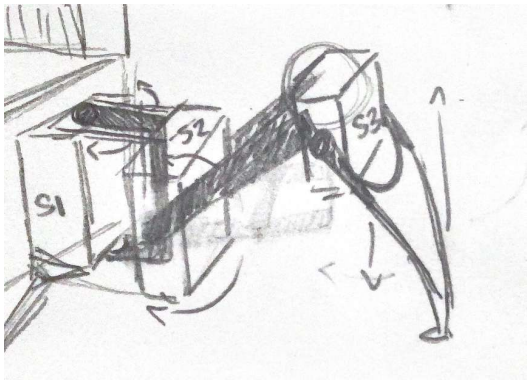
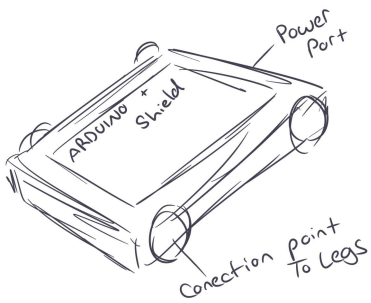
Appendix A: Definitions, Acronyms, Abbreviations

HMC: Howl's Moving Castle

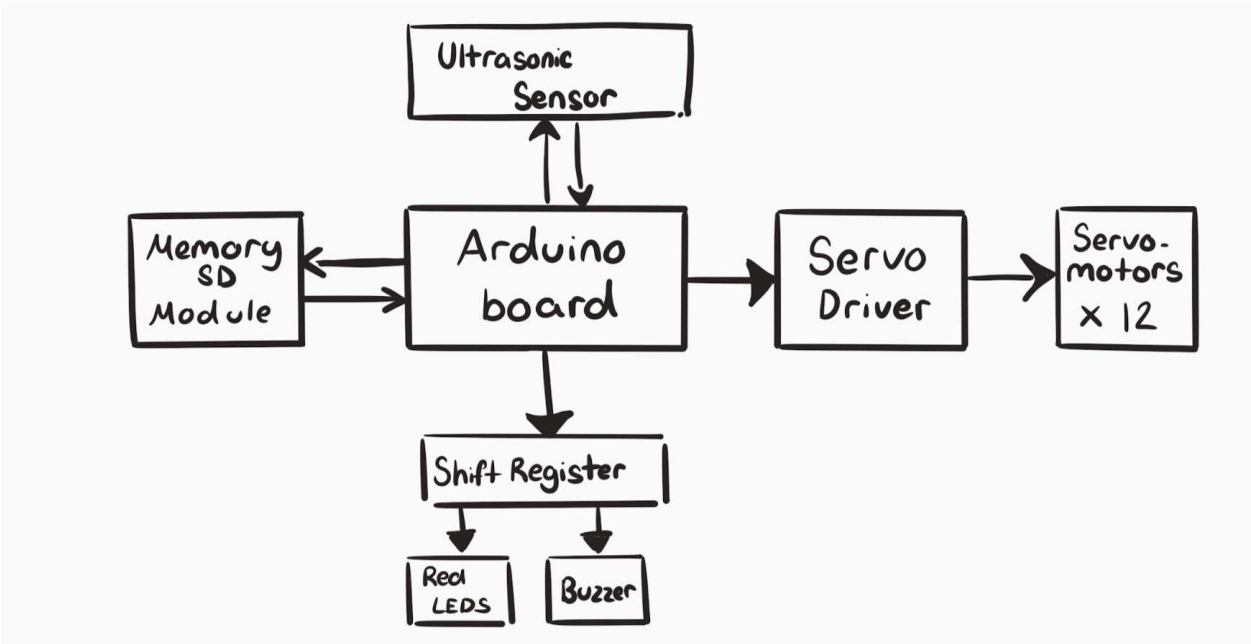
PWM: Pulse With Modulation

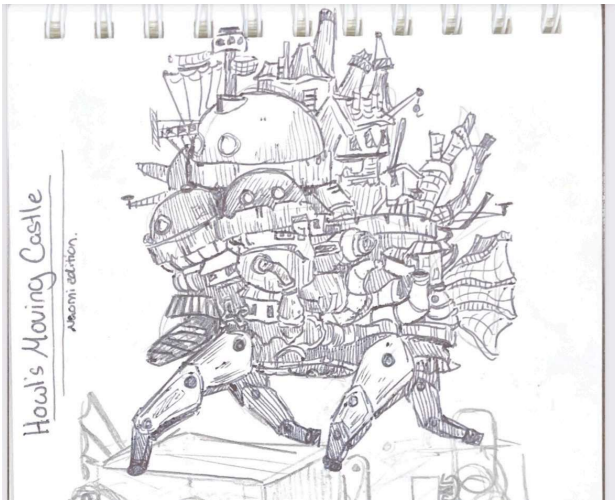
OST: Original Sound Track

The Castle	The Body

	
The Legs	The Base
	

Appendix B: Concept Art and Photos





Appendix C: Design Inspiration

Robot quadrupeds:

- https://www.instructables.com/QUATTRO-the-Arduino-Quadruped-Robot/
- https://robotlk.com/3-in-1-spider-robot-quadruped-using-arduino/
- https://www.youtube.com/watch?v=fvUhFBq7Z4g
- https://youtu.be/sDuzb-I00-4?si=4OrCCQTd9GRdIOER

Castle:

- https://youtu.be/iwROgK94zcM?si=OZ8UwEC4zgZ2lMkj 0:0 → 0:12, 0:50 →0:55
- https://youtu.be/FDQ-sDDqWvk?si=X_2DNbg1nvXuihIH
- https://youtu.be/iDCEm_mSQnU?si=vE7ONwlVLQO2EYaR